

INVESTIGATION 8.2

Drag strips

AIM To analyse the motion of an accelerating object using ticker tape

Materials:

ticker timer
G-clamp
power supply
ticker tape (in 60 cm lengths)
toy car (or dynamics trolley)
sticky tape or masking tape
clear length of bench at least 60 cm long

METHOD AND RESULTS

- ▶ Clamp the ticker timer firmly to the edge of a table or bench so that you will be able to pull 60 cm of ticker tape through it. Connect the ticker timer to the AC terminals of the power supply and set the voltage as instructed by your teacher.
- ▶ Thread one end of the ticker tape through the ticker timer so that it goes under the carbon paper disc.
- ▶ Attach the end of the ticker tape to the toy car or trolley.
- ▶ Turn on the power supply and check that the ticker timer leaves a black mark on the ticker tape.
- ▶ Model a drag-racing car by pushing the toy car or trolley forward, starting from rest, so that it reaches a maximum speed near the halfway mark. Make it come to a gradual stop near the end of the 'track'.
- ▶ Remove the ticker tape and mark off the first clear dot and every fifth dot after the first. Each interval between the marks represents a time of $\frac{5}{50}$ of a second; that is,

0.1 seconds. Measure the distance travelled during each 0.1 second interval and write it on your tape. Label the intervals as interval 1, interval 2, interval 3 and so on.

- 1 Construct a table like the one below in which to record your data. Calculate the average speed during each interval and record it in the table.
- 2 Now cut your ticker tape into 0.1 second intervals and use the strips as described in Investigation 8.1 to construct a graph of speed versus time.

DISCUSS AND EXPLAIN

- 3 Describe the motion of the toy car or trolley during the period over which it was recorded. Ensure that the words *speed*, *accelerated* and *decelerated* are used in your description.
- 4 Between which intervals was the acceleration:
(a) positive (b) negative?
- 5 During which interval did the greatest average speed occur?
- 6 When did the greatest positive acceleration take place?

Drag strip speeds

Interval	Distance travelled (cm)	Average speed (cm/s)
Example	3.6	$\frac{3.6 \text{ cm}}{0.1 \text{ s}} = 36$
1		
2		
3		

UNDERSTANDING AND INQUIRING

REMEMBER

- 1 Why is the accelerator in a car called by this name?
- 2 Explain the difference between acceleration and deceleration.

CALCULATE

- 3 A car that has stopped at a set of traffic lights sets off when the lights turn green. It increases its speed by 5 m/s during each of the first 3 s after it sets off, and by 3 m/s during the following 2 s.
 - (a) What is the speed of the car after:
 - (i) 1 s
 - (ii) 2 s
 - (iii) 5 s?
 - (b) What is the average acceleration of the car during the first 5 s after it sets off?

PROCESS AND ANALYSE

- 4 Repeat Investigation 8.2 using data-logging equipment and a motion detector or light gates. Use the software to produce a graph of speed versus time. Print a hard copy of your graph.
 - (a) Use coloured pencils or a highlighter pen to indicate the part of the graph that shows the car or trolley speeding up. Use a different colour to indicate the part of the graph that shows the car or trolley slowing down.
 - (b) What is the maximum speed of the car or trolley?
 - (c) At what time was the maximum speed reached?
 - (d) How would your graph be different if the trolley sped up more quickly, yet reached the same maximum speed?
 - (e) Calculate the average acceleration of the car or trolley used in your own experiment.

eBookplus

work
sheet

8.3 Acceleration